11 There are p counters in a bag.
12 of the counters are yellow.

Shafiq takes at random 30 counters from the bag.
5 of these 30 counters are yellow.

Work out an estimate for the value of p.

Yellow proportion in sample = 5/30 = 1/6.

Estimate 12/p = 1/6, so p = 12 × 6 = 72.

2 marks total
Sample proportion
Estimated total 72

.....

(Total for Question 11 is 2 marks)

12 $T = \frac{q}{2} + 5$

Here is Spencer's method to make q the subject of the formula.

$2 \times T = q + 5$

$q = 2T - 5$

What mistake did Spencer make in the first line of his method?

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(Total for Question 12 is 1 mark)

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